

RENANCE

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# FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MEE 213 — Past Questions & Materials  
**by Renance**

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Course: MEE 213  
Institution: Federal University of Petroleum Resources, Effurun  
Papers in this pack: 6  
Levels: 200

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## Paper 1 of 6 — MEE 213(2018/ 2019)

Level 200 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN  
COLLEGE OF TECHNOLOGY  
DEPARTMENT OF MECHANICAL ENGINEERING  
FIRST SEMESTER EXAMINATION 2018/2019 SESSION  
COURSE CODE AND TITLE: MEE 213 – BASIC MANUFACTURING TECHNOLOGY

Ged Eng Rec Qu

INSTRUCTION: ANSWER ANY FOUR (4) QUESTIONS; TIME ALLOTTED: 2Hrs

**Question 1**

- (a) What is manufacturing?
- (b) An external threading operation needs to be performed at one end of a solid metal shaft measuring 720mm (length) and 60mm in diameter. List the tools/machines and also the step by step procedure required to carry out this operation.
- (c) Briefly explain (i) cold working processes (ii) hot working processes (iii) engineering workshop. **(17.5 marks)**

**Question 2**

- (a) (i) List and explain any three (3) workshop layout known to you. (ii) List three (3) advantages of a good workshop layout
- (b) During an interview section in a ship building company, the candidates were asked to suggest with reasons the steel/material which they would recommend for building the body of a ship and various answers were given. Assuming you were in that interview section what would have been your answer?
- (c) List, describe and sketch any three (3) fitting tool available in the workshop **(17.5 marks)**

**Question 3**

- (a) (i) Define the term "hazards" as it relates to workplace (ii) List any three (3) sources of hazards known to you in the workshop.
- (b) Briefly explain these common workplace hazards (i) manual handling operations (ii) slips, trips and falls (iii) fatigue among workers (iv) chemical hazards
- (c) Write short notes on (i) extrusion drawing (ii) wire drawing (iii) hot drawing. **(17.5 marks)**

**Question 4**

- (a) Write short notes on the following drilling operations (i) drilling (ii) reaming (iii) boring (iv) counter-boring (v) counter-sinking (vi) spot facing
- (b) A mild steel was firmly held to a bench vice for a drilling operation. If it is required to drill a hole of 12mm diameter in the mild steel of 40mm thick, when the cutting speed of the drill is 28mm/min. How long will this machining operation last if the feed of the drilling machine is 0.15mm/rev? **(17.5 marks)**

**Question 5**

- (a) (i) Define the term accident as it relates to workplace. (ii) Accidents be it minor or major may occur in a workplace; name three (3) common causes of its occurrence in the workplace.
- (b) As a workshop trainee in fupre mechanical workshop, you are required to drill a 60mm diameter hole at the centre of a metal plate measuring 160mm (diameter) x 12mm (thick). List the tools/machines and step by step procedure you will require to carry out this operation.
- (c) List and describe any two (2) measuring or marking tool you know. **(17.5 marks)**

**Question 6**

- (a) Basically, manufacturing involves three (3) fundamental steps; name them
- (b) Manufacturing processes is generally classified into six major groups; name them and explain briefly any three of your choice.
- (c) Describe briefly any five (5) available metal known to you that is use in sheet metal works in the workshop. **(17.5 marks)**

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## Paper 2 of 6 — MEE 213(2015 2016)

Level 200 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES,

\_ EIFFURUN

Ne DEPARTMENT OF MECHANICAL ENGINEERING \_

: Resende First Semester Examination (2016/2016'Session) ~ ~ . |

i. COURSE CODE/TITLE: MEE 213 - BASIC MANUFACTURING

gs TECHNOLOGY

TIME ALLOWED: 23 hours:

INSTRUCTION: ANSWER QUESTION 1 AND ey OTHER THREE

### QUESTION 1

(a) G) What is plant layout? @i) With the aid of a detailed sketch show the layout of asheet metal and painting shop (iii) Write briefly on any four (4) types of fire extinguishers (iv) Take the vernier caliper and micrometer readings as shown in figures 1 and 2 respectively. i 10 marks

(by) @) Discuss four (4) out of the major groups in-which manufacturing processes are classified Gi) With the aid of a neatly labeled sketches distinguish between acetylene and oxygen cylinders. (ii) List and briefly explain the various plant layouts : : 15'marks

### QUESTION 2

(@). Shortly define the following: (i) Welding (ii) Gas and are welding processes (iii). ~. - welding rods and electrodes (iv) Soldering and brazing (v) Resistance, welding

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~ py @) What is a stake and stake holder Gi) Write short notes on metals used in sheet metal work . . 18 : Bae i \_ Fmarks

### QUESTION 3

(a) @) Define accidents and state their effects on fret

ory owner, workers and society.

(ii) Briefly. explain the various types of fire

7 marks —

(b) @ Explain the various types of flame used for gas welding (ii) In a resistanée ~ welding operation, voltage bf 223V was supplied to 'allow tt ¥

, OL ; ne passage electric current through

each work piece of 4.3kQ. resistance for 12.5 minutes, Calculate the amount of heat genetated in each work piece. os

8 marks -

### QUESTION 4 ~ ee

(a) (i) State five (5) characteristics of good Riahand Gi) Distin:

hard wood with examples for each (iii) oe ie ees

a List the various carpentry joints Z7marks.

: (b), @) Distinguish between casting and pattern makin a. a

commonly used for pattern making (iii) Following the b

g Gi) State th af aterials

asic steps in met ti

production, explain how you will cast a hollowed object (i a

#### QUESTION 5

.apipe) 8 marks

(a) Briefly write on the following engineering materials: (i) Metals and /

Ferrous and non-ferrous metals (iii) Cast iron and steels 6 mar

ot

+e"

(b) Gi) Write on the two types of milling methods (ii) With the aid of a neat'sketch '

show the motions of a twist drill (ii) Briefly explain the various most common eh

operations which can be performed on a drilling machine 9 marks

#### QUESTION 6

(a) @) What is a lathe? (ii) With the aid of a neat sketch briefly explain the feed \_  
mechanism of a lathe , 7 marks :

(b) G@) With the aid of a neat sketch explain ary three (8) most common operations

which can be carried out on a lathe (ii) How long will it take to dril. a hole of 0.5cm

diameter in a mild steel plate of 58mm thickness when the cutting speed of drill is

45m/min, and feed is 0.26mm/rev. . ' ; & marks

## Paper 3 of 6 — MEE 213(2017 2018)

Level 200 • First Semester • PastQuestion

DEPARTMENT OF MECHANICAL ENGINEERING, COLLEGE OF TECHNOLOGY, FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN, DELTA STATE NIGERIA.

2017/2018 FIRST SEMESTER EXAMINATIONS

COURSE CODE: MEE 243 ,

COURSE TITLE: BASIC MANUFACTURING TECHNOLOGY

INSTRUCTIONS: Answer question one and any other two

questions, Orderliness and good logic

in the presentation of facts will be rewarded. :

Question One:

A. Write short notes on the following: (i) Idle speed of an internal combustion engine (ii) Burn in test (iii) work area and work space (iv) counter boring (v) reaming and (vi) drilling.

B. A road side mechanic recommended eight (8) - cylinder engine block to power a mini refinery that requires a 3600cm<sup>3</sup> capacity engine in Warri. The length of stroke of each cylindrical bore selected was 16.0 cm and the diameter of the bore was measured to be 9.0 cm. Estimate the actual engine capacity of this block and advise the mechanic on the implication of his decisions in the engineering world:

C. What do you understand by the term industrial hazards? How can we eliminate it in our work places?

Question two

A. With the aid of a well labelled diagram, discuss the configurations and uses of the following workshop tools; (i) the hammer (ii) the double cut file (iii) the hacksaw. and (iv) the Centre Punch, :

List and explain some ways in which workshop tools can be maintained for optimal performance. :

Question three

A. There are three fundamental steps involved in all manufacturing processes, identify these steps and write short notes on them

B. Write short notes on the qualities of a good manufacturing process in our industrial setting.

Question four : °

A. A mild steel was firmly held by a bench vice for a drilling operation, if the thickness of the mild steel is 62.0 mm, and it is required to drill a hole of 7.0 mm diameter in the mild steel. How long will it take this machine to accomplish this task if, the cutting speed of the drill is 32 m/ min and the feed of cut of the drilling machine is 0.16 mm/ rev. or as a

Name two each of the following fitting tools found in our workshops and briefly explain their uses; (i) cutting tools (ii) drilling tools (iii) measuring tools (iv) My striking tools and (v) holding tools, : :

Question five

A. With regards to the advantages derived from good workshop Organization and its latent objectives, explain what you understand by workshop organization.

B. Write short notes on the following; (1) welding and soldering (ii) soft wood and hard wood (iii) processes effecting changes in engineering properties,

## Paper 4 of 6 — MEE 213 PQ (2022-2023) Past Question

Level 200 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN  
DEPARTMENT OF MECHANICAL ENGINEERING  
First Semester Examination, 2022/2023 Session  
MEE 213- BASIC MANUFACTURING TECHNOLOGY  
Time Allowed: 2 hours

Instruction: Answer four questions in all. Question No. 1 is compulsory

**QUESTION 1**

Part A:  
List five different machine tools used in a workshop and briefly describe their main functions.

Part B:

- Explain the working principle of a lathe machine.
- Provide a labelled sketch of a lathe machine, showing its major components and how they work together during operation.
- Discuss two common applications of lathe machines in manufacturing processes.

**QUESTION 2**

(a) Match each manufacturing process in Row A with its corresponding application in Column B.

Row A: Casting, Machining, Forming, Joining, Additive manufacturing

Column B:

- Production of complex geometric shapes with high accuracy and surface finish
- Creation of custom tooling, prototypes, and low-volume parts
- Production of large and complex shapes with little material waste
- Assembly of two or more parts into a single component
- Production of hollow or solid parts with a wide range of sizes and shapes

(b) Describe the main steps involved in the casting process. Provide an example of a product that is commonly produced using casting.

**QUESTION 3**

- Define the following: (i) Industrial safety (ii) industrial hygiene (iii) accident (iv) workshop hazard (v) first aid
- Discuss the various types of workshop hazards.
- Discuss any five safety protective wears.

**QUESTION 4**

- Define the following: (i) Computer Aided Engineering (ii) Computer Aided Manufacturing.
- What are the Benefits and Roles of a computer?
- What is input and output device?

**QUESTION 5**

- What is manufacturing technology?
- Explain the various basic Manufacturing processes
- Discuss the general principles of manufacture that will ensure the profitability of manufacturing organization.

## Paper 5 of 6 — MEE 213 PAST QUESTIONS

Level 200 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES,

EFFURUN

- DEPARTMENT OF MECHANICAL ENGINEERING.

First Semester Examination (2015/2016 Session)

COURSE CODE/TITLE: MEE 213 - BASIC MANUFACTURING

TECHNOLOGY

TIME ALLOWED: 2} hours

INSTRUCTION: ANSWER QUESTION 1 AND ANY OTHER THREE

### QUESTION 1

(a) @) What is plant layout? (ii) With the aid of a detailed sketch show the layout of a sheet metal and painting shop (iii) Write briefly on any four (4) types of fire extinguishers (iv) Take the vernier caliper and micrometer readings as shown in figures 1 and 2 respectively. 10 marks

(6) @) Discuss four (4) out of the major groups in which manufacturing processes are classified (ii) With the aid of a neatly labeled sketches distinguish between. acetylene and oxygen cylinders. (iii) List and briefly explain the various plant . layouts : 15 marks —

### QUESTION 2

(a) Shortly define the following: @) Welding (ii) Gas and arc welding processes (iii) welding rods and electrodes (iv) Soldering and brazing (v) Resistance welding & marks

(b) G) What is a stake and stake holder Gi) Write short notes on metals used in sheet metal work 7 marks

### VU QUESTION 3

(a). @) Define accidents and state their effects on factory owner, workers and society.

(ii) Briefly explain the various types of fire 7 marks

(b) @ Explain the various types of flame used for gas welding (ii) In a@ resistance welding operation, voltage of 223V was supplied to allow the passage electric current through each work piece of 4.3kQ resistance for 12.5 minutes, Calculate the amount of heat generated in each work piece. 8 marks

### QUESTION 4

(a) @) State five (5) characteristics of good timbers (ii) Distinguish between soft and, hard wood with examples for each Gii) List the various carpentry joints 7 marks

(b) @) Distinguish between casting and pattern making (ii) State the materials commonly used for pattern making (iii) Following the basic steps in metal casting production, explain how you will cast a hollowed object (i.e. a pipe) 8 marks

## QUESTION 5

(a) Briefly write on the following engineering materials: (i) Metals and Alloys (ii) Ferrous and non-ferrous metals (iii) Cast iron and steels 6 marks

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DEPARTMENT OF MECHANICAL ENGINEERING we a ee:

FIRST SEMESTER 2014/15 SESSION ; .

213 BASIC MANUFACTURING TECHNOLOGY

MEE

TIME: 1hr 30 mins

ATTEMPT ANY FOUR QUESTIONS

## QUESTION ONE

(a) What is Plant Layout? List and explain the various kinds of plant layout (10 vw marks). °.

e holder? (ii) Mention five (5) different types of

(b) | What isa stake and stak

stake with the aid of good sketches (15 marks). |—

## QUESTION TWO

(a) What is an acci

© marks).

(b) Take the micrometer and Vernier caliper

and 2 respectively (15 marks} | e

dent? List and explain five (5) types of accidents (10

readings as shown in figures 1.

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\_ Figure 1: Micrometer

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Foxe 2: Vernier caliper— '> = (is) uy b a4 b&

Bote aca Sal => ss

| a) pe wen

| =° \* athe 42mm

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ee

a fire? Mention seven (7) major Industrial \~

### QUESTION THREE

(a) Mention and explain three (3)

safety objectives? (10 marks)

(b) Neatly produce labeled sketches of amemplleme and oxygen cylinders (15 marks). ;

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eae

### QUESTION FOUR ae

(a) List the various types of drilling macivime, Draw and label a twist drill (245 marks)

(by) In a resistance welding operation, current of 22mA was passed through each work piece of 10kQ resistance for 10 minutes Galelate the amount of heat generated in

" each work piece (10 marks) a care # :

ve ue port y ee

### QUESTION FIVE / R20 ae

(a) Briefly explain the following: (i) Brazing (i) foldering (ii) Fluxes (iv) Welding rods (16 marks). < 3 :

(b) Distinguish between soft and hardwood @5 marks). ma is

### QUESTION SIX 3 Po aes oe

i (a) List and discuss the variou hinds of molidling send available (10 marks). '

: (2) What is welding? List six (6) classification af welding processes (5 marks) Vv

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DEPARTMENT OF MECHANICAL fiesta

First Semester Examinetion, 2022/2023 session

MEE 213- BASIC MANUFACTURING TECHNOLOGY

Time Allowed: 2 hours

Instruction: Answer four questions in all. Question No, 1 is compulsory

### QUESTION 1

Part A:

List five different machine tools used in a workshop and briefly describe their main functions,

Part B:

- (i) Explain the working principle of a lathe machine,
- (ii) Provide a labelled sketch of a lathe machine, showing its major components and how they work during operation, together
- (iii) Discuss two common applications of lathe machines in manufacturing processes.

### QUESTION 2

(a) Match each manufacturing process in Row A with its corresponding application in Column B.

Row A: Casting, Machining, Forming, Joining, Additive manufacturing

Column B:

- (i). Production of 'complex geometric shapes with high accuracy and surface finish
  - (ii). Creation of custom tooling, prototypes, and low-volume parts
  - (iii). Production of large and complex shapes with little material waste
  - (iv). Assembly of two or more parts into a single component\*
  - (v). Production of hollow or solid parts with a wide range of sizes and shapes
- (b) Describe the main steps involved in the casting process, Provide an example of a product that is commonly produced using casting.

### QUESTION 3

(a) Define the following: (i) Industrial safety (ii) industrial hygiene (iii) accident (iv) workshop hazard (v) first aid

(b) Discuss the various types of workshop hazards.

(c) Discuss any five safety protective wears.

### QUESTION 4

(a) Define the following: (i) Computer Aided Engineering (ii) Computer Aided Manufacturing,

(b) What are the Benefits and Roles of a computer? /

(c) What is input and output device? - }

### QUESTION 5

9. What is manufacturing technology?

: Explain the various basic Manufacturing processes

: Discuss t

88h zation eed principles of manufacture that will ensure the profitability of ~4.,

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COLLEGE OF TECHNOLOGY

DEPARTMENT OF ii CHANTCAL ENGINEERING V4

First Semester Examination (2016/2017 Session) .

COURSE CODE/TITLE: MEE 213 - BASIC MANUFACTURING TECHNOLOGY

TIME ALLOWED: 2hours 15 minutes

INSTRUCTION: ANSWER TWO QUESTIONS FROM EACH SECTION

SECTION A \y

QUESTION 1 yy)

ring precesses (b) List and explain the basic production

ated with appropriate production methods. (c) Briefly

(i) Accidents (iii) Hazards (iv) Protective Wears

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN ‘

(a) Explain the various basic manufactul

methods. Hence, state the layout associ

explain the following: (i) Engineering Workshop

(v) Sheet metal work

QUESTION 2 :

(a) Distinguish by definition and example between marking out and fitting (b) List, sketch and describe any five marking-out tools. (c) List, sketch and describe any five fitting tools.

QUESTION 3

(a) (i) Distinguish between carpentry ahi

With the aid of a neat sketch, distinguis

of diagrams the following basic types of we

Joint (iv) Lap Joint (v) Edge Joint

d joinery (ii) List five auxiliary tools used in carpentry Gin

h between butt weld and fillet weld (b) Explain with the aid

\ded joints: (i) Butt Joint (ii) Tee Joint (iii) Corner

J - oe ”

SECTION B

QUESTION 4 ett

(a) (i) What is welding? Write sho!

Distinguish between soldering and b!

Oxy-acetylene welding (iii) Filler rods (iv

rt notes on the three types of oxy-acetylene welding flames (ii)  
 razing (b) Write short notes on the following: (i) Arc welding (ii)  
 ) Fluxes (v) Welding electrodes

#### QUESTION 5

(a) (i) List five advantages of timber (ii) State the chara'  
 seven carpentry joints (b) (i) What are measuring instru  
 measuring instruments you know (give examples for each)  
 cteristics of soft and hard woods (iii) List  
 ments? (ii) List and explain the classes of

#### QUESTION 6

ween pattern and casting (ii) List and explain five pattern materials (iii) List  
 (a) (i) Distinguish bet  
 taken in casting a hollow object.  
 ten foundry hand tools (iv) List and explain the basic steps  
 i  
 pot mas

## Paper 6 of 6 — MEE 213 HANDOUT

Level 200 • First Semester • Handout

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### MEE 213 – BASIC MANUFACTURING TECHNOLOGY

#### Course Outline

1. Introduction to basic manufacturing processes
2. Organisation of workshop
3. Workshop hazard
4. Safety practice and code
5. Properties of engineering materials
6. Bench-work and fitting
7. Introduction to turning
  - Straight and step turning
  - Chamfering
  - Screw cutting
8. Milling
9. Drilling techniques
10. Sheet metal work
11. Welding, brazing and soldering technique
12. Foundry techniques
13. Properties of wood
14. Woodwork and joinery
15. Workshop measurements

1

### CHAPTER 1-INTRODUCTION TO BASIC MANUFACTURING PROCESSES

#### 1.0 Outline of Manufacturing Process

Manufacturing process is that part of the production process which is directly concerned with the change of form or dimensions of the part being produced.

It does not include the transportation, handling or storage of parts, as these are not directly concerned with the changes into the form or dimensions of the part produced.

By way of elucidation, production process itself denotes the process followed in a plant for converting semi-finished products or raw materials into finished products. More succinctly, manufacturing process is the art of converting raw materials into finished goods with the application of different types of tools, equipment, machine tools, manufacturing setups and manufacturing processes.

For simplicity, manufacturing process will be canvassed under two broad headings in this course, namely:

- (i) Primary shaping process, and
- (ii) Secondary or machining process

We shall take them seriatim.

(i) Primary Shaping Process

These include the following:

- casting
- powder metallurgy
- Plastic technology
- Gas cutting
- Bending, and
- Forging(cold and hot processes)

2

(ii) Secondary or Machining Process

In this dimension, we mean the following processes:

- Turning
- Threading
- Knurling
- Milling
- Drilling
- Boring
- Planing
- Shaping
- Slotting
- Sawing
- Broaching
- Hobbing
- Grinding
- Gear cutting
- Thread cutting, and
- Unconventional machining processes.

Besides, under this category, we also have

- Computer numerical control(CNC)
- Tooling using:
  - LBM
  - ECM
  - AJM
  - USM Setups

The foregoing listings have helped to give a vista of known production (manufacturing) processes. In sections of text that follow, we shall briefly and precisely discuss these processes. However, before we do that, let us examine some historical background of manufacturing processes.

3

### 1.1 Historical Background of Manufacturing processes

Borrowing from the language of the bible, in the beginning, that is in ancient times dating back to about 9500 BC, at around North-east of the present Iraq, metals were used as found, or at best hot-forged to the extent that the then available manufacturing process allowed. This crude process was used in manufacturing copper pendants. A pendant is an ornament on a chain worn around the neck.

Progress in primary shaping process took place in Mesopotamia in 3000 BC with the evolution of technology of copper smithing. Also in Egypt the lost wax casting process evolved. Later, the Chinese introduced bronze casting and did extensive work in this area and the period became known as bronze age. On account of contiguity factor, the Japanese developed on casting. All these took place in the Middle-and Far- East. Almost contemporaneously progress in casting process was recorded in the West too. For instance, a bronze statue casting weighing 360 tons, towering 105 feet high was developed in Rhodes near Greece. It was called the colossus of Rhodes, which was set at the entrance to the harbour of ancient Rhodes in 285 BC. It is acknowledged as one of the Seven Wonders of the World. For the purpose of completeness, it is instructive to note that Rhodes is the largest island of the Dodecanese group: 545 square miles. It is the chief city, capital of the Dodecanese Islands.

Moreover, in Belgium at Marche-Les-Dames in 1340 BC, casting process technology had attained continuous volume production of molten iron .And in 1826 BC in England, Black-heart casting process for malleable iron was developed by Seth Boyden. Around this time also, Henry Sorby developed the metallurgy science. Spectacularly in 1948, shortly after the end of the World war II, nodular cast iron controls namely:

- Castability of cast iron, and
- Impact strength of steel, were introduced.

4

### 1.2 Brief Treatment of Primary Shaping Process

#### 1.2.1 Casting

The word casting has double meaning.

1. Casting is a manufacturing process involving melting metal, pouring it into mould cavity and allowing the molten metal to solidify as a casting (final product of casting process) whose shape is a reproduction of the mould cavity.
2. Casting also means the product resulting from a casting process.

Having defined the pun, we shall proceed by offering definitions of some important foundry man terms.

#### 1. Cupola furnace

It is employed for smelting metal or pig iron for production of various cast irons. It is also used for production of nodular and malleable cast iron.

#### 2. Refractories

This is ability of moulding sand to withstand high temperatures without breaking down or fusing thus facilitating the process to get sound casting. It is an important characteristic of moulding sands.

### 3. Permeability

This is known as porosity of the moulding sand in order to allow the escape of any air, gases or moisture present or generated in the mould when the molten metal is poured into it. All these gases generated during pouring and solidification process must escape otherwise casting becomes defective. Permeability is a function of grain size, grain shape, and moisture and clay contents in the moulding sand. The extent of ramming of the sand directly affects the permeability of the mould.

5

### 4. Pattern

This denotes the model or the replica of the object(to be cast). It should be emphasized that a pattern prepares a mould cavity for the purpose of making a casting.

Fig. 1: Mould Making

### 5. Core prints

Core is a solid form placed in a mould about which molten metal is poured, so as to cast hollow. Put in another way, core is part of the mould that forms desired internal cavity, recess or projection in casting.

Core print is a projection added to both sides of the pattern to create impression that allows the core to be supported and held at both ends.

### 6. Mould cavity

This refers to the void space or hollowness allowed in a mould.

### 7. Core boxes

This simply represents a mould for producing cores which after casting can create hollow in the cast product.

Core boxes are generally made of wood or metal and are of several types. The main types of corebox are half core box, dump core box, split core box, strickle core box, right-and left- hand core box and loose piece core box.

6

### 8. Shrinkage allowance

This is excess dimensions accommodated in the exact dimensions of cores to compensate for reduction in size that takes place during cooling of casting.

### 9. Cope and drag

This refers to upper and lower segments of a pattern respectively.

See figure 23 for illustration.

### 10. Chaplets

Fig.2

Fig.3: use of chaplets

These are metal distance pieces inserted in a mould either to prevent shifting of mould or locate core surfaces. (see figure 2 and 3)

The distance pieces in form of chaplets are made of parent metal of which the casting is. Chaplets are placed in mould cavity suitably so as to adequately position core and to give extra support to core and mould surfaces. Different types of chaplets exist – radiator chaplet, riveted chaplet, welded chaplet, double head chaplet, cast chaplet, sheet metal chaplet etc.

7

Chills

These are fast heat extracting metallic material incorporated into sand mould in order to produce a hard part of casting by making that portion to cool faster than the rest. Chills are of different types and shapes.

Fig.4: Types of Chills

Fig.5: Use of a Chill

Foundry Tools and Equipment

Figures (6) – (21) refer. They represent different tools and equipment commonly used in foundry work.

Fig.6: Hand riddle

8

Fig.7(b): Rammers

Fig.7(a): Shovel

Fig.9: Strike off bar

Fig.8: Sprue pin

Fig.11: Vent rod

Fig.10: Draw spike

Fig.12: Lifters

Fig.13: Trowels

9

Fig.14: Slicks

Fig.15: Smoothers

Fig.16: Swab

Fig.19: Open moulding boxes

Fig.18: Bellows

Fig.17: Gate cutter

Fig.20: Closed rectangular moulding boxes

Fig.21: Closed round moulding box

10

### Casting processes

Under this consideration, we shall note the following.

1. Metal material in form of scraps or native form of a metal(ore) is brought to molten state by heating in a furnace such as:

- Refractory crucible
- Iron(for lead), or
- Refractory-lined furnace structures

Fig(22): elements of melting system

2. Heat supply is usually by any of the following:

- electric
- gas
- oil heating
- magnetic induction
- mixing the charge itself with the fuel, e.g coke, in blast furnace or cupola

3. The charge is made to yield the alloy of the desired composition upon complete melting.

4. Usually a protective atmosphere is an important requirement. It could be any of the following:

- air (with associated humidity and various pollutant)
- a protective atmosphere (e.g argon,nitrogen etc.)
- a vacuum

11

5. The charge is mixed with silicate of various metal oxides or fluxes (inorganic compounds) which can form slag that float on top of the molten metal as a protective layer.

6. The pouring process in its simple form is achieved with the aid of cope-and-drag sand mould whose scheme is illustrated in Fig (23) that follows.

Fig.23: Characteristics of a cope-and-drag sand mould

Molten metal is poured into pouring basin. It runs down the tapered sprue down to the well where impurities settle while pure molten metal flows through the runner, passing the ingate to the mould cavity and core. The molten metal in the feeder head assures that voids are not created in the casting.

### Castings

Once the pattern(s) and core(s) have been produced, the sand mould is made in the foundry.

The mould is made in a moulding box or 'flask' comprising an upper box or 'cope' and a lower box or 'drag'. Dowels and sockets align the two halves which are usually made either of steel, cast iron or of aluminium alloy.

12

Fig(24): The flask

The moulding sand is either the traditional 'greensand' which is a mixture of sand grains and clay particles, such as Mansfield or Erith sands, or now more commonly, clean sand with oils or synthetic resins added as binders. The desired properties are a compromise between permeability and smoothness coupled with cohesion. The moisture content should be such that a handful remains unbroken after gripping firmly, but loose enough to break when thrown down. Over-dampness may cause bubbling and a porous finish to the casting.

#### Four stages in Moulding and Pouring

##### Stage 1

The drag is placed upside down on a firm flat surface. If a split pattern is used, the half with sockets is placed face down and then dusted with a parting powder. (Note: detail is usually better in the bottom half of the castings, as gas and dross will rise.) Handfuls of sifted sand are then thrown at the pattern, effectively covering every detail; this is known as 'sand slinging'. The sand is then lightly rammed either with a small peg rammer or the finger tips. Loose unsifted sand is shovelled in and firmed with a butt rammer. The excess is strickled off and a handful sprinkled over and rubbed across to ensure a flat surface.

13

Fig(25): Stage 1: Moulding

##### Stage 2

The drag is then turned over in one smooth movement. The cope is fitted on to the drag. To prevent the two halves of sand sticking, a parting powder is used, which can be a proprietary make, french chalk or more conveniently the burnt sand brushed from castings. The powder can be dusted from a cotton bag.

Sprue pins, or 'runner sticks', which are smooth round tapered pegs, are pressed firmly into the sand. The position is determined by considering the flow of metal and the convenience of excess metal removal afterwards ('fettling').

The cope is crammed with sand and strickled off, making the top smooth and firm. If a large amount of air will have to be driven through the sand, vent holes should be made within about 10mm of the pattern. A pouring basin or sprue cup is cut before removing the sprue pins.

14

Fig(26): Stage 2: Moulding

##### Stage 3

The cope is then carefully removed. A depression or skim bob is often scooped out at the base of the pouring gate, to trap particles and to reduce turbulence. A runner or channel is then cut to link the pouring gate with the mould; this should be slightly smaller than the sprue diameter, to prevent air inclusion and turbulence. One or more fairly large runners join the mould to the riser or feeder head (Fig. 28)

The sand around the edge of the two halves of patterns is moistened to prevent crumbling. A spike or screw is driven into the pattern. This spike is then rapped in all directions before the pattern is lifted vertically.

Loose sand must be removed, for it is better to cast too much metal rather than too little. For a smooth finish the mould can be lightly coated with plumbago or a proprietary product.

The core(s) is (are) now placed in position and the cope carefully replaced , ready for pouring.

15

Fig(27): Stage 3: Moulding

Stage 4

A metal such as aluminium can be melted in a crucible; it is first fluxed to give protection from the atmosphere and to separate metal from dross, and then degassed of hydrogen by plunging a proprietary tablet to the bottom of the melt.

The best means of temperature control is to use a special foundry thermometer protected by a sheath.

The crucible is held about half- way down with special tongs and withdrawn from the furnace. Dross is raked away from the pouring lip with a heated skimmer and the metal poured in one continuous stream until it appears at the head of the riser.

Usually the thickest parts show the greatest shrinkage, therefore either a large feeder head is placed nearby or a sleeve of combustible material is placed at the top of the riser to keep the head fluid until the casting has solidified.

16

Fig(28): Stage 4: Pouring

To harden the surface of cast iron, metal chill plates are inserted into the mould.

When the casting has solidified and cooled, the sand is knocked out and the casting fettled.

Die Casting

The fundamental principle of this method is that metallic moulds are used rather than sand. Molten alloys of aluminium, copper, lead, magnesium, tin and zinc are forced by pressure or gravity into preheated moulds. The high cost of the dies limits the process to the mass production of accurate and well-finished small -to- medium castings.

17